INTRODUCTION TO DATA SCIENCE

Module 1 — Notes Summary

Compiled from my hand-written class notes

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1. Foundations of Data Science

1.1 What Is Data Science?

- **Data Science** is the field of exploring, manipulating and analyzing data, and using it to make recommendations and informed decisions.
- It is the **study of large quantities of data** which can reveal insights that help organizations make strategic choices.
- Not a brand-new field — data analysis existed before, but it has **evolved**: more, varied data sources (logs, sensors, social etc.) demand bigger computing power.
- Driven by **Big Data + Computing Power**: modern tech enables analysis of huge datasets efficiently.

The study of data to understand world

1.2 Why Organizations Need Data Science

- Help organizations **understand their environment**, **solve problems**, and **find hidden opportunities**.
- Good data scientists are: **Curious**, **Argumentative**, **Judgmental**.

Who is DS? Who finds solns to problems by analyzing data using appropriate tools.
Delivering knowledge & insights from data.

- Flow: **Data → Stories → Insights** — clarify the problem, collect data, analyze, tell the story, and visualize it.

1.3 The Data Science Process (5 Steps)

Step	What Happens
1. Define the Problem	Clarify exactly what the organization wants answered (the most important step).
2. Data Collection	Identify what data is needed and where to get it (structured & unstructured).
3. Analysis & Modeling	Use multiple models/algorithms to find patterns, trends, outliers, new insights.
4. Insight Generation	Results may confirm assumptions or reveal completely new knowledge.
5. Communication	Present findings clearly to stakeholders so they can act on them.

1.4 Key Terms

- **Algorithm**: a step-by-step set of instructions to solve a problem.
- **Model**: a simplified version / representation of relationships & patterns found in data, used to make predictions.

- **Outliers**: data points that are very different from the rest.
- **Quantitative analysis:** approach to analyze numerical data using mathematics and statistics.
- **Structured data**: data is organized, formatted into predictable schema — easy to analyze.
- **Unstructured data**: unorganized data that makes it harder to analyze using traditional methods, which includes image, text, videos.

Analytical Skill:- Thinking logically with data to solve problems

1.5 File Formats Used in Data Science

Working with data means working with many different file formats, and it helps to know the structure, benefits, and limits of each one:

- **Delimited text files (CSV, TSV)**: **Simple data storage** - values separated by commas or tabs; the first row is usually the column names. Easy to process and widely supported. TSV is useful when comma itself appears in original data.
- **XLSX (Excel format)**: **Spreadsheet Operation** - spreadsheet format with rows, columns, and cells, able to hold multiple worksheets; based on XML and widely supported.
- **XML [Extensible Markup Language]: Structured data sharing**- stores data using custom, nested tags; readable by humans and machines, and platform/language independent.
- **PDF [Portable Document Format]**: **Fixed Doc display -** a fixed-layout format that displays documents consistently on any device; common for legal, financial, and form documents.
- **JSON [Java Script Object Notation]**: **Web & API data exchange** - a lightweight, text-based format for structured data, used heavily in APIs and web services.

Data file type: (container) The kind of file used to store data. eg: CSV, excel, JSON.
Data format: how data is organised inside it. The way data is arranged inside the file. eg: rows & columns, key-value pairs, tags.

Data Science is study of large quantities of data which can reveal insights, that help organizations strategic choices.
Many algorithms are used to bring out insights from data

2. Cloud Computing for Data Science

2.1 What Is Cloud Computing?

- Delivery of **on-demand computing resources** (networks, servers, storage, applications, services, data centers) - the physical or virtual components of a computer system, hosted and made available over the internet.
- Lets users access **applications and data** over the internet rather than locally (e.g. web-based apps, online business apps, cloud storage like Google Drive / Dropbox).
- Billed on a **pay-per-use basis**.

2.2 Benefits

- No need to install applications on local computers.
- Use the online versions of the applications; pay a monthly subscription instead.
- More cost-effective; saves local storage space.
- Enables **real-time collaboration** on shared work.

2.3 Essential Characteristics

- **On-demand self-service** — access resources anytime, without human intervention.
- **Broad network access** — usable from any device (mobile, laptop).
- **Resource pooling** — shared resources serve multiple users efficiently.
- **Rapid elasticity** — scale up/down as needed.
- **Measured service** — pay only for what you use.

2.4 Deployment Models

How cloud resources and services are made available to users.

Model	Description
Public Cloud	Shared services over the internet, available to anyone.
Private Cloud	Used exclusively by one organization.
Hybrid Cloud	Combination of public + private cloud.

2.5 Service Models

Model	What It Provides
IaaS	Provides servers, storage, networking infrastructure.
PaaS	Provides tools to build and deploy applications.
SaaS	Provides software via subscription (ready-to-use apps).

3. Big Data & Data Mining

Digital Transformation

Digital transformation means using digital technology to change how an organization operates and delivers value.

Big data is a major driver of this shift - companies that use data effectively can improve performance and stay competitive

simple :-
use data + technology to rethink and improve
everything in an organization.

3.1 Big Data Foundations

- In the digital world, everyone leaves a **digital footprint**.
- **Big Data** = dynamic, large and disparate volumes of data generated by people, tools and machines.
- Requires new, innovative and scalable tech to **collect, host and analytically process** the vast amount of data gathered to derive real-time business insights.

3.2 The 5 V's of Big Data

V	Meaning
Velocity	Speed of data generation.
Volume	Large amount of data.
Variety	Diverse data types (text, video, etc.).
Veracity	Data quality / accuracy.
Value	Useful insights derived from data.

- Traditional tools aren't enough — need tools like **Apache Hadoop** and **Apache Spark**.

Goal: Turn data into useful insights and decisions.

3.3 Data Mining

- A **repeatable, step-by-step process** to turn raw data into useful insights for decision-making.

Step	Description
1. Goal Set	Identify the key aim.
2. Select / collect Data	Identify the data sources.

Step	Description
3. Preprocess	Clean the data.
4. Transform	Transform data by determining the appropriate format to store the data.
5. Data Mine	Determine methods & execute.
6. Evaluate	Assess outcomes, share results.

- Data mining is **iterative** — repeat and improve.

3.4 Impact of Big Data & Cloud computing

- Drives **digital transformation** in business and society.
- Help organizations improve: **decision-making**, **performance**, **customer understanding**.
- Big data tools:
  - **Apache Hadoop** – Storage and distributed processing
  - **Apache Hive** – Query and analyze data.
  - **Apache Spark**: Fast data processing
- **Distributed data**: dividing data into smaller chunks & processing them across multiple computers within a cluster — enables parallel processing for faster analysis.
- Cloud computing makes handling big data **easier and more cost-effective**.

Big data + cloud + Data Mining = better insights & smarter decisions.

4. AI, Generative AI & Data Science

4.1 AI vs Data Science vs ML

- **Machine Learning (ML)**: algorithms that learn from data and make predictions without explicit programming.
- **Data Science**: broad field that extracts insights from data using math, statistics, and ML.
- **AI**: focuses on making machines intelligent.
- **DS**: focuses on analyzing data for insights — both fields use big data & ML.

4.2 Generative AI (Gen AI)

- A subset of AI focused on **producing new data**, rather than just analyzing existing data.
- Creates images, text, video, code, etc.
- **GAN** and **VAE** are foundation for this tech— these models replicate the underlying features of the original data.

4.3 Generative AI in Data Science

- Creates **synthetic data** when real data is limited — mimics real data patterns/distributions.
- Helps **improve model training** by adding more data.
- **Automates coding** for building analytical models — saves time and effort.
- **Generates insights and reports** automatically as data updates / changes.
- Can **discover hidden patterns** in data.
- Some tech use natural-language AI to analyze data.
- **Gen AI helps you work faster, smarter, and with much larger language models.**

5. Applications & Careers in Data Science

5.1 How Companies Get Started with Data

"If they are unable to measure something
they are unable to improve it."

- If a company is unable to **measure their goals**, they are unable to achieve them.
- If unable to **measure profit**, unable to increase it.

> ***"The first thing a company has to do is start capturing data."***

- Once you have data, you can apply **algorithms & analytics** to it.
- If you are capturing data, **archive it** — data never gets old; data is always relevant (even 10, 20, 50 years from now).
- **Do not overwrite** old data — thinking you don't need it anymore is a common mistake.
- Put best practices for data archiving in place **the moment** you start a business.

5.2 Applications of Data Science

Application	Example
Recommendation Systems	Amazon, Netflix, Spotify — suggest products based on past behaviour.
Personal Assistants	AI uses data science to answer user queries.
Location-Based Services	Location based recommendations.

5.3 How Organizations Use Data Science

- **Drive business goals**, improve efficiency, make predictions.
- Example — Healthcare (EHR): analyze patient data to discover **patterns in diseases**.
- Transform massive data into **actionable, life-saving insights**.

All organizations use DS for same reason:

Discover optimal solutions to existing problems.

Find new insights for further improvement.

5.4 Case Studies

Uber

- Use data to put the **correct number of drivers** in the right place at the right time.

Streaming Company

- Collects mass data from millions of users — which shows watched, what time of day, pause/rewind/fast-forward.
- As a result, the company **confidently predicts the success of a show before filming begins**, by thoroughly analyzing user-preference behaviors.

Healthcare

- **Predictive analytics** helps find the best treatment options for patients.
- DS helps doctors recommend appropriate tests early and suggest treatments.
- Developing more extensive data analytics capabilities supports healthcare is moving from **descriptive analytics → predictive analytics**.

5.5 Presenting Findings (Final Deliverable)

- When presenting findings to your organization: **explain conclusions**; offer them in the form of a research paper if appropriate; establish your narrative.
- Aim for: **Powerful**, **Convincing**, **Evidence-based and well-written**.

5.6 Technical Skills

- **Python programming, Statistics, Probability, Calculus** — "problem-solving abilities & analyzing problems" (curiosity).
- Retail-oriented environment → **Python, statistical skills**; if big data → **Hadoop, Spark**.
- Communication Skills.

Main reasons for the emergence of vast amounts of data

IoT + **distributed computing** — brought vast amounts of data and the technological capability to analyze it.

5.7 Report Structure

> ***"Good analysis is not enough — a well-structured and clearly written report is essential to communicate insights effectively."***

Type	Description
Brief Report	Short, direct findings.
Detailed Report	Full research-paper style: Coverage, TOC, summary, introduction, literature review, methodology, results, discussion, conclusion, references, appendix (if needed).

- **Checklist for good writing**: clear purpose & aim, show practical usefulness, suggest future work, maintain logical structure.

7. Data Literacy — Types & Sources of Data

7.1 Understanding Data

- **Data**: raw, unorganized information — includes facts, numbers, text, symbols.
- Becomes **meaningful after processing and analysis**.

7.2 Types of Data (Based on Structure)

Type	Key Features	Examples
Structured	Well-defined schema; organized in rows & columns (tables); easy to store, query & analyze.	SQL DB, spreadsheets, online forms, sensors (GPS, RFID), web server logs
Semi-structured	Has some structure but no fixed schema; uses tags / metadata; organized in hierarchical format (not tabular).	XML, JSON, zipped files
Unstructured	Heterogeneous, no fixed structure; cannot be stored in tables easily; highly diverse & complex.	Social media posts, images, videos, audio, PDFs, documents, web pages

7.3 Data Sources

- Today's data comes from **many different sources**, both inside and outside organizations — diverse, dynamic, and continuously growing.

Internal Data Sources (within the organization)

- Most companies store their data in **relational databases** — managed using **MySQL**, **Oracle DB**, **Microsoft SQL Server**.
- Examples: sales transactions, customer CRM data, employee/HR systems.
- Used for: sales analysis, forecasting, business decision-making.

External Data Sources (outside the organization)

- Organizations also use outside data to improve decisions: government datasets (population, economy), purchased data (weather, financial).
- Formats: flat files (simple csv files), spreadsheets, XML files.

7.4 XML (Semi-Structured Data)

- Data stored using **tags** supports **hierarchical structure** (e.g. bank statements, survey data).
- More flexible than flat files — shows hierarchical structure.

7.5 APIs & Web Services

- **API**: a way for applications to request and receive data.

- How it works: send request → receive response (JSON / XML).
- Use cases: **Twitter, Facebook APIs** (social media analysis); **stock market APIs** (stock prediction); **data validation APIs** (data cleaning).
- Very important for **real-time and automated data access**.

7.6 Web Scraping

- Extracting data directly from websites.
- Tools: **Beautiful Soup, Scrapy, Selenium**.
- Can extract: text, images, product prices, reviews.
- Use cases: price comparison, dataset creation (collecting training data for ML models).
- Used when **APIs are not available**.

7.7 Data Streams (Real-Time Data)

- **Data streams** = continuous flow of data in real time.
- **Source:** IoT devices, GPS systems, social media, Sensors
- **Example:** Stock market updates, Website clicks, Industrial sensors
- **Tools**: **Apache Kafka, Apache Spark Streaming** — used for real-time analytics and decision-making.

7.8 RSS Feeds

- **RSS** = Really Simple Syndication — automatically captures updated data from online forums & news sites where data is refreshed on an ongoing basis.
- A feed reader collects updates and shows latest content — useful for tracking updates continuously.

7.9 Why Data Sources Matter

Source Type	Examples
Internal	Databases
External	Files, APIs
Web-based	Scraping, feeds
Real-time	Streams

- Data scientists need to: **collect data from multiple sources**, **handle different formats**, and **work with structured (databases), semi-structured (XML/JSON) and unstructured (web, media) data**.

8. Metadata & Metadata Management

8.1 What Is Metadata?

- **Metadata** = data about data.
- Provides info describing other data — helps users understand: **what** the data is, **where** it comes from, **how** to use it.

8.2 Types of Metadata

Type	Describes
Technical	Data structure (table names, columns/rows, data types).
Process	Properties of data, source and time — helps in troubleshooting.
Business	Describes data in simple terms — helps business users understand the data easily.

Example — student database table:

Technical: column metadata describes field names (Stud_ID, Name, Age, Course, Marks) and data types.

Process: Data loaded on DD-MM-YYYY, last updated by admin.

Business: Meaning of data for users.

8.3 Why Metadata Matters

- Imagine a search system: you search "Marks of Data Science students." Metadata helps the system find the right **table** (Students) and **column** (Marks).
- **Without metadata it is very difficult to locate data.**

8.4 Metadata Management

- The practice of **organizing, storing and controlling** metadata.

Benefit	Why It Matters
Data Discovery	Easily find relevant data.
Repeatability	Reuse data and processes.
Data Governance	Ensures data quality, security, availability.
Data Lineage	Tracks data's origin and flow.

- **Analog (older) records** may exist in formats like paper
- **Catalog** - "digital format" represents data in non-physical, machine-readable form. Refers to information such as text, images, video – encoded into binary code (0 &1) that computers can process, store, and transmit.

9. Data Collection and Organization

9.1 Data Collection & Organization

- **Data repository**: a general term used to refer to data that has been collected, organized and isolated so it can be used for business operations, reporting & analysis.
- Simply: data repositories (like databases, data warehouses and big data stores) are systems used to **store, manage and analyze data efficiently**.

9.2 Databases & DBMS

- **Database**: a collection of data stored and accessed in computer system; supports storage, retrieval, modification.
- **DBMS**: software used to manage databases — allows querying (finding specific data), e.g. find active customers using queries.

9.3 Types of DB: Relational vs Non-Relational Databases

	Relational (RDBMS)	Non-Relational (NoSQL)
Structure	Data stored in tables (rows & columns)	Flexible, schema-less
Schema	Fixed schema	Handles unstructured / big data
Used for	SQL, structured data	Social media, IoT, cloud-based systems

- Choice of DB depends on: **data type & structure**, querying mech, **transaction** speed.

9.4 Relational Databases — Structure

- Data is organized into **tables** (rows & columns).
- **Rows = records**, **Columns = attributes**.
- Tables are linked/connected using common data fields, e.g. **Customer table ↔ Transaction table**, linked using a common field such as Customer ID.
- **Key concept - Relationships**: tables can be joined using common data, enabling creation of new insights and reports (e.g. customer statements).

9.5 Advantages of Relational Databases

- **Efficient** for large data (faster than spreadsheets);
- **Reduced redundancy** (data stored once, not duplicated).
- **Data integrity & consistency** (data types and constraints visa keys).
- **High performance** (process millions of records quickly).
- **security & access control**.
- **Easy backup & recovery**.
- **ACID compliance**.

9.6 Types & Use Cases of Relational Databases

Category	Examples
Open Source	MySQL, PostgreSQL
Commercial	Oracle DB, Microsoft SQL Server, IBM DB2
Cloud-based (DB as a Service)	Amazon RDS, Google Cloud SQL, SQL Azure

- **Common use cases**:
    - **OLTP** (online transaction processing — fast transactions, insert/update/delete);
    - **Data Warehousing** (OLAP, historical analysis);
    - **IoT applications** (handling data from edge devices).

9.7 Limitations of Relational Databases

- Not suitable for **unstructured or semi-structured data**.
- **Schema must match** — migration is hard.
- **Fixed field size limits**;
- **less ideal for big data analytics**.

Final summary: Despite limitations, relational DBs remain dominant for structured data.

10. NoSQL Databases

10.1 What Is NoSQL?

- "**Not only SQL**" — a non-relational DB with flexible schema.
- Supports structured and unstructured data; popular in **big data, cloud computing & modern web/mobile apps**.

10.2 Key Characteristics

- **No fixed schema** — flexible data structure.
- Does not rely on traditional tables (rows/columns);
- often does not use SQL (uses SQL-like queries).
- Designed for **scalability & performance** and **ease of use**.

10.3 Types of NoSQL Databases

Type	How Data Is Stored	Use Cases	Examples
Key-Value Store	Key-value pairs — fast & simple	Session data / caching	Redis, Memcached, Amazon DynamoDB
Document-Based	Stored as documents (e.g. JSON) - flexible queries & indexing	E-commerce, CRM, medical records	MongoDB, Couch DB, Amazon DocumentDB
Column-Based	Store in columns instead of rows — high performance for large-scale data	Time-series data, IoT analytics	Cassandra, Apache HBase
Graph Database	Stored as nodes (entities) and edges (relationships)	Social networks, fraud detection, recommendations	Neo4j, Azure cosmos DB.

10.4 Advantages of NoSQL

- Handles **large volumes of diverse data**;
- **highly scalable** (distributed across multiple servers).
- **Cost-effective** (runs on commodity hardware).
- **flexible & agile** development.
- **Better suited for modern applications**.

10.5 RDBMS vs NoSQL — Comparison

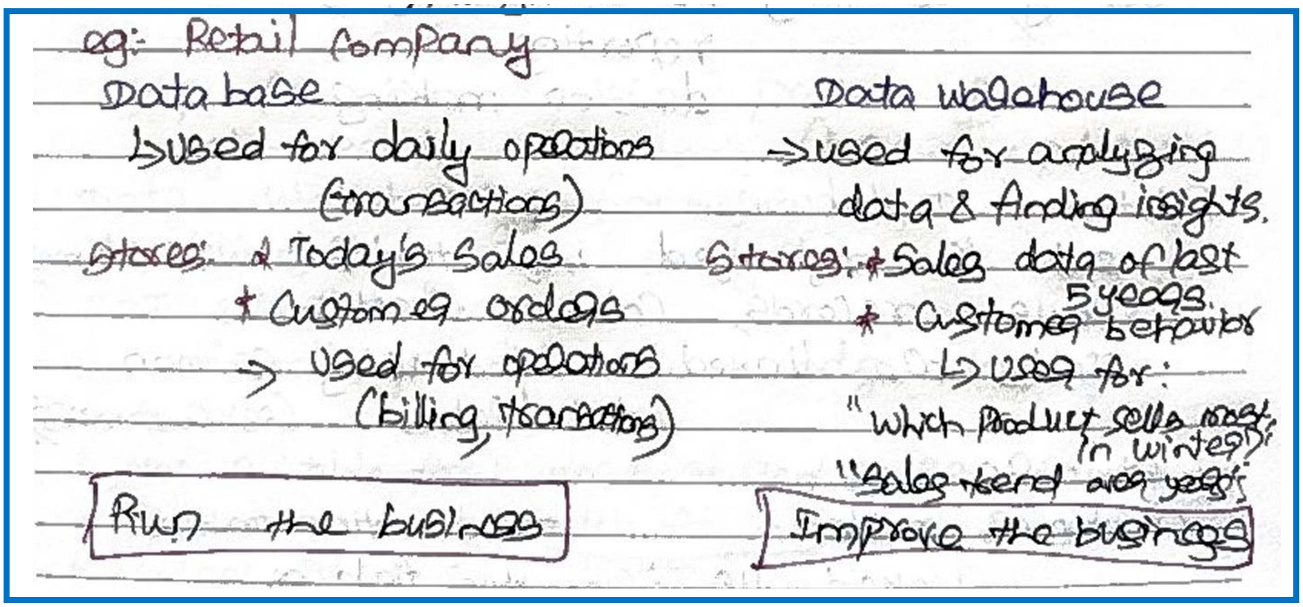
	RDBMS	NoSQL
Schema	Fixed schema	Flexible / schema-less
Data Type	Structured only	Structured + unstructured
Cost	Often expensive	More cost-effective
Transactions	ACID-compliant	Often not fully ACID

- NoSQL emerged to **overcome limitations of RDBMS** — increasingly used in large-scale, real-time, and data-intensive applications.
- **Both NoSQL and RDBMS are important, depending on the use case.**

11. Data Warehouses, Marts, Lakes & Pipelines

11.1 Data Warehouse

- One of the **data repositories**, like databases — large companies (data-driven organizations) mainly use data warehouses.
- **Central system** that: a central repository that merges info coming from disparate sources.
- Consolidates data through the **ETL process** into one comprehensive database for analytics and business intelligence.
- Stores **processed, structured, and analysis-ready** data;
- Contains current + historical data.
- Data is cleaned, transformed & organized.
- Used for **reporting & business analytics**; best for organizations with large operational data needing analysis.



eg:- Retail company

Data base
↳ used for daily operations (transactions)
Stores: * Today's Sales
* Customer orders
→ Used for operations (billing, transactions)
Run the business

Data warehouse
→ used for analyzing data & finding insights.
Stores: * Sales data of last 5 years.
* Customer behavior
↳ Used for:
"Which product sells most in winter?"
"Sales trend over years"
Improve the business

11.2 Data Mart

- A **subset of a data warehouse** — designed for specific business functions (e.g. Sales, Finance).
- Provides **targeted data access**, **faster performance**, and **better security (isolated)**.
- Main use: department-specific reporting & analytics.
- Smaller, focused part of a data warehouse.

11.3 Data Lake

- Stores **raw data in its native format** — supports structured, semi-structured and unstructured data. Data is tagged with metadata.
- **Key feature**: no predefined schema; stores all data without filtering.
- Used for: **big data storage**.

11.4 Data Warehouse vs Data Lake

	Data Lake	Data Warehouse
Data state	Raw	Processed
Structure	Flexible	Structured
Purpose	All data types	Purpose-specific

11.5 ETL Process (Extract, Transform, Load)

- Convert raw data into analysis-ready data.

Stage	What Happens	Methods / Tools
1. Extract	Collect data from sources.	Batch processing (scheduled, large chunks) - Stitch, Blendo; Stream processing (real-time data) - Apache Kafka, Storm, Samza
2. Transform	Clean & prepare data: standardize formats, remove duplicates, filter unnecessary data.	—
3. Load	Move processed data into the target system.	Initial load, Incremental load, Full refresh

11.6 Data Pipelines

- Covers the **entire data flow** — source, processing, destination.
- **End-to-end** process of collecting, transforming and moving data between systems.
- One common pipeline flow: **Data Source → Data Lake (optional) → ETL/ELT → Data Warehouse → Data Mart ->Analysis**.

11.7 Choosing the Right Data Repository

- Considerations before choosing: am I using it more for **transaction processing**, or for **analytical** purposes?
- Also weigh **security features** and **scalability**.
- Before you pick the right DB, evaluate **nature of applications**, **volume of the data**, **structure of data**.